

INJECTION MOULDING

From polymer granules to millions of identical parts — the machine, the tool, the materials, the defects and every design rule on one sheet

Billions	of identical parts/made every year
2–60 s	full moulding/cycle time
7,000+ t	clamp force on/the largest presses
±0.01–0.05 mm	typical/tolerance
g → 50+ kg	shot size — from/micro to jumbo parts

THE PROCESS WHAT IS INJECTION MOULDING?

Injection moulding is the mass-production process for plastics. Polymer **granules** are melted inside a heated barrel, then forced under very **high pressure** through a nozzle into a **closed steel mould**. The melt fills the hollow **cavity**, cools and solidifies to the exact cavity shape, and the mould opens to eject the finished part. Each cycle takes only **seconds**, so the machine makes **identical parts at very high rate** — the billions of bottle caps, LEGO bricks, phone cases and medical syringes we use every day.

How it differs from other plastic processes

- vs **blow moulding** — solid 3-D parts, not hollow sealed bottles
- vs **extrusion** — discrete parts, not endless profiles
- vs **thermoforming** — fully 3-D detail, not thin shells
- Only economical above ~10,000 parts — the tool pays for itself

THE TOOL ANATOMY OF THE MOULD (DIE)

- Core & cavity** — male block core + female hollow cavity; together form every surface of the part.
- Parting line** — the meeting plane of the two halves; the part must release from it.
- Ejector system** — **pins / sleeves / stripper plate** push the cooled part off the core.
- Cooling channels** — water lines (often conformal) remove heat fast; cooling dominates cycle time.
- Sprue / runner / gate** — the feed system carrying melt to the cavity.
- Vents** — 0.02–0.05 mm grooves along the parting line that let trapped air escape (prevent burn marks).
- Slides / lifters** — moving tool parts that form **undercuts** (side holes, barbs, internal lips).

Mould materials

- P20** — pre-hardened general-purpose steel (most moulds)
- H13** — hot-work steel for long runs, abrasive/glass-filled resins
- S136** — stainless, polishable mirror finish (optics, medical)
- Beryllium copper inserts** — high thermal conductivity for hot spots

A tool costs £10k–£250k+, survives 500k–several million shots, and is machined to mirror-image accuracy of the part.

MACHINE ANATOMY THE INJECTION MOULDING MACHINE

Four main assemblies work together:

- Injection unit** — melts the plastic and pumps it in: **barrel** (heated cylinder), **reciprocating screw**, **band heaters** around the barrel, and the **nozzle** sealing the barrel to the mould.
- Clamping unit** — **platens** (fixed + moving steel plates) and a **toggle/hydraulic clamp** that shut the mould and hold it locked, rated by **clamp force** in tonnes/kN (up to 7,000+ t).
- Mould / die** — the precision steel tool that shapes the part.
- Control system** — PLC sets temperature, pressures, speeds and timings for every shot.

The 4 temperature zones along the barrel

- Feed zone** — cool end; granules arrive from the hopper
- Compression zone** — granules melt as depth reduces
- Metering zone** — melt mixed to uniform temperature
- Nozzle** — delivers the melt into the sprue bush

WHICH MACHINE? MOULDING MACHINE TYPES

By drive technology

- Hydraulic** — powerful, proven; large clamp forces; higher energy use
- Electric (all-electric)** — servo motors on every axis; fast, precise, repeatable, up to 60% less energy; premium machines
- Hybrid** — electric screw/injection + hydraulic clamp; best of both

By clamping mechanism

- Toggle clamp** — mechanical linkage locks with little oil; fast open/close
- Hydraulic clamp** — straight cylinder; force easy to control; common on large presses

Rated by **clamp force** — the maximum tonnes/kN holding the mould shut (30 t to 7,000+ t). Too little = flash; too much = wasted energy.

Shot size vs clamp — the 50–70% rule

Aim to use **50–70% of the rated shot (injection) capacity** so the plastic melts consistently in the barrel; very large or very small parts relative to the press idle it or degrade the resin.

THE SCREW & BARREL PLASTICISING — GRANULES TO MELT

Correct screw design plasticises (melts) the resin in three steps:

- 1. Feed zone** — granules conveyed forward, still solid
- 2. Compression zone** — screw depth tapers; friction + band heaters melt the granules
- 3. Metering zone** — constant-depth section mixes the melt smooth and uniform

As the screw **rotates** it melts, mixes and pushes the melt to the front, then pulls back — this is **screw recovery** (plasticising), overlapping the cooling phase to save time.

The screw as a **plunger**: at injection the screw stops rotating and is **rams forward like a plunger**, forcing the melt through the nozzle, sprue, runners and gate into the cavity at up to 1,500 bar.

L/D ratio 15–25:1 · compression ratio ~2–3:1 · **back pressure** (resistance on screw return, 5–20 bar) improves mixing & melt uniformity.

ONE CYCLE THE MOULDING CYCLE — 7 STEPS

- 1. Close** — moving platen slides the mould shut; clamp locks it under ton-force
- 2. Inject** — screw rams forward, forcing melt into the cavity (1–3 s)
- 3. Pack / hold** — extra pressure (50–80% of injection) squeezes in more melt to feed shrinkage
- 4. Cool** — melt solidifies against the chilled steel; the longest phase (10–40 s+)
- 5. Open** — clamp pulls the moving half away; part sticks to the core
- 6. Eject** — pins/sleeves/stripper push the part off; robot or gravity removes it
- 7. Repeat** — screw recovers the next shot during cooling, so the cycle restarts immediately

Total cycle 20–60 s (fast micro parts under 2 s). **Clamping force needed** = cavity pressure × projected part area — the force trying to blow the mould open, which the clamp must exceed. Cooling time dominates (~50–70% of the cycle), so a well-cooled mould is the fastest mould.

RUNNER SYSTEMS HOT RUNNER vs COLD RUNNER & GATING

How the melt is delivered to the cavity — and the trade-off between scrap, speed and tool cost.

Feature	Cold runner (conventional)	Hot runner	Ref
Runner waste	Sprue + runners + gates freeze and are trimmed off	Melt stays molten inside the tool — near-zero scrap	waste
Cycle time	Longer — runner must cool and eject	Faster — no runner to cool or eject	cycle
Tool cost	Lower, simpler mould	Much higher — heated manifolds, valve gates, more controls	cost
De-gating	Manual or semi-auto trimming after ejection	Part comes out gated clean — automatic	gating
Typical use	Low-medium volumes, cheaper tools, any material	High volume, thin-wall, precise or multi-cavity parts	use
Gating types	Edge, tunnel/submarine, fan, sprue, diaphragm	Pin (point), hot tip, valve gate	gates

SET THE PROCESS KEY PARAMETERS

Every setting lives in a **process window**. Move one variable, re-shoot, verify. Semi-crystalline resins (PP, PA, POM) shrink most; amorphous (ABS, PC, PS) least.

Parameter	Typical value	Effect & note
Melt temp	180–320°C	PP 220–250 · ABS 220–260 · PC 280–320°C; too cold = short shot, too hot = degraded resin
Mould temp	20–120°C	PP 30–60 · ABS 40–80 · PC 80–120°C; set by cooling channels, affects gloss/warpage
Injection pressure	up to 1,500 bar	Forces melt into every cavity detail; too high risks flash
Hold (pack) pressure	50–80% of inj.	Held 1–2 s while gate open to feed shrinkage; prevents sink marks
Cooling time	10–40 s+	Dominates the cycle; set by wall thickness & mould temp
Screw speed	50–250 rpm	Faster = shorter recovery but more shear heating
Back pressure	5–20 bar	Resistance to screw return; improves mixing, less air trapping

WHICH POLYMER? MATERIALS & SHRINKAGE

Moulds are cut for ONE resin — choose the polymer before the tool. Shrinkage is allowed for by cutting the cavity oversize.

Polymer	Melt °C	Mould °C	Shrink %	Typical parts
PP (polypropylene)	220–250	30–60	1–2.5	caps, crates, living hinges, bumpers
HDPE	190–230	20–60	2–3.5	bottles, buckets, fuel tanks, toys
ABS	220–260	40–80	0.4–0.7	LEGO, housings, dashboards, phones
PS (polystyrene)	190–250	20–60	0.4–0.8	cutlery, CD cases, model kits
PC (polycarbonate)	280–320	80–120	0.5–0.7	goggles, headlamps, lenses
PA (nylon)	240–300	40–90	0.7–1.5	gears, bearings, clips, under-bonnet
POM (acetal)	190–220	60–100	1.5–2.5	precision gears, pump parts, sliders
PVC	170–200	20–60	0.2–0.6	fittings, cable plugs, soles
PET (polyester)	260–295	110–140	1.5–2.0	bottle preforms, trays, jars
PMMA (acrylic)	220–260	50–90	0.3–0.7	lenses, light covers, tail-lights

TRADE-OFF ADVANTAGES vs LIMITATIONS

✓ **ADVANTAGES**

- Complex 3-D shapes with internal detail in one shot
- Extremely fast cycles — up to thousands of parts/hour
- Moulded-in finish, precision & repeatability (LEGO ±0.002 mm)
- Moulded-in features: threads, inserts, living hinges
- Near-zero waste — sprues/runners re-ground as regrind
- Very low labour cost per part once running

✗ **LIMITATIONS**

- Very high tooling cost (£10k–£250k+)
- Only economical for high volume (>~10,000 parts)
- Weeks of mould design + machining lead time
- Changes are costly once the tool is cut
- Part size limited by machine shot capacity
- Undercuts & tight tolerances push cost up sharply

QUALITY CONTROL DEFECTS — RECOGNISE, CAUSE, FIX

Every defect has a cause in the process window or the design. Change ONE variable at a time, log the process sheet, re-shoot.

Defect	What it looks like	Main cause	Prevention / fix
Short shot	Part incomplete; cavity not fully filled	Not enough material, low pressure, cold melt, gate too small	Increase shot, pressure, melt/mould temp; enlarge gate
Flash	Thin plastic burr at the parting line	Clamp force too low; worn or ill-fitting mould; vent gap too big	Raise tonnage; repair/reset mould; lower injection pressure
Sink marks	Shallow depressions on thick sections	Local shrinkage at thick walls; hold pressure released early	Uniform walls or ribs; hold longer; cool longer
Weld lines	Faint lines where melt fronts meet	Two flow fronts don't fully fuse (around holes/inserts)	Raise temps; move/add gates; add vents
Burn marks	Black/brown scorch at end of fill	Trapped air 'diesels' (burns); melt too hot	Vent last-filled areas; lower melt temp; slow injection
Warpage	Part twists or bows after ejection	Uneven shrinkage from uneven cooling/wall thickness	Balance cooling; uniform walls; symmetric gates
Voids	Air pockets inside thick walls	Trapped air or internal shrinkage	Thinner walls; higher hold pressure; better venting
Jetting	Squiggly worm-like streaks from gate	Melt shoots across cavity, freezes before fill	Raise temps; larger gate; ramp injection speed up
Splay	Silver streaks / moisture marks	Damp pellets or moisture flashing to steam	Dry the resin; lower melt temp; check venting

WHERE IT'S USED APPLICATIONS

If it's plastic, identical and made by the million, it was almost certainly injection moulded.

- Automotive** — bumpers, dashboards, interior trim, air vents, light housings
- Electronics** — phone & laptop housings, connectors, switches, casings
- Medical** — syringes, IV connectors, inhalers, device housings (cleanroom)
- Consumer** — toys (LEGO), bottle caps, pens, kitchenware, power tools
- Packaging** — caps, closures, trays, jars, food containers, crates

Blow moulding takes over for hollow sealed shapes (bottles, tanks); **extrusion** for endless profiles.

VOCABULARY GLOSSARY — KEY TERMS

Injection unit — melts and pumps the plastic

Screw — rotating helix that melts, mixes, injects

Nozzle — seals barrel to mould, delivers melt

Cavity — female hollow forming the outside

Runner — channels feeding melt to each cavity

Sprue — main entry channel from the nozzle

Parting line — meeting plane of the two halves

Shrinkage — polymer contraction cooling (0.2–3.5%)

Clamping unit — holds the mould shut under clamp force

Barrel — heated cylinder containing the screw

Mould / die — precision steel tool that shapes the part

Core — male block forming the inside/hollow

Gate — restricted entry into the cavity

Ejector — pins/sleeves pushing the part out

Cooling channels — water lines removing heat

Cycle time — total time for one complete shot

QUICK REFERENCE CORE FACTS & RECALL

The exam line: 'describe how a product is injection moulded' → 1 close & clamp the mould, 2 screw rams melt into the cavity, 3 hold pressure while it cools & shrinks, 4 open the mould, 5 eject the part, 6 trim the runner/gate.

Thermoplastic vs thermoset

- Thermoplastic** — cool mould re-solidifies it; repeatable in seconds; runner scrap re-ground & reused. Standard injection moulding.
- Thermoset / elastomer** — hot mould cures by cross-linking; cannot be re-melted.

Cooling dominates the cycle · clamp force = cavity pressure × area · 50–70% shot rule · gates at the thickest section · draft 0.5–2° · undercuts = slides/lifters.